

Tramacastilla_disturbance_palaeoecology_analyses

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This is the code used to perform statistical analyses included in Julián-Posada et al. (in prep).

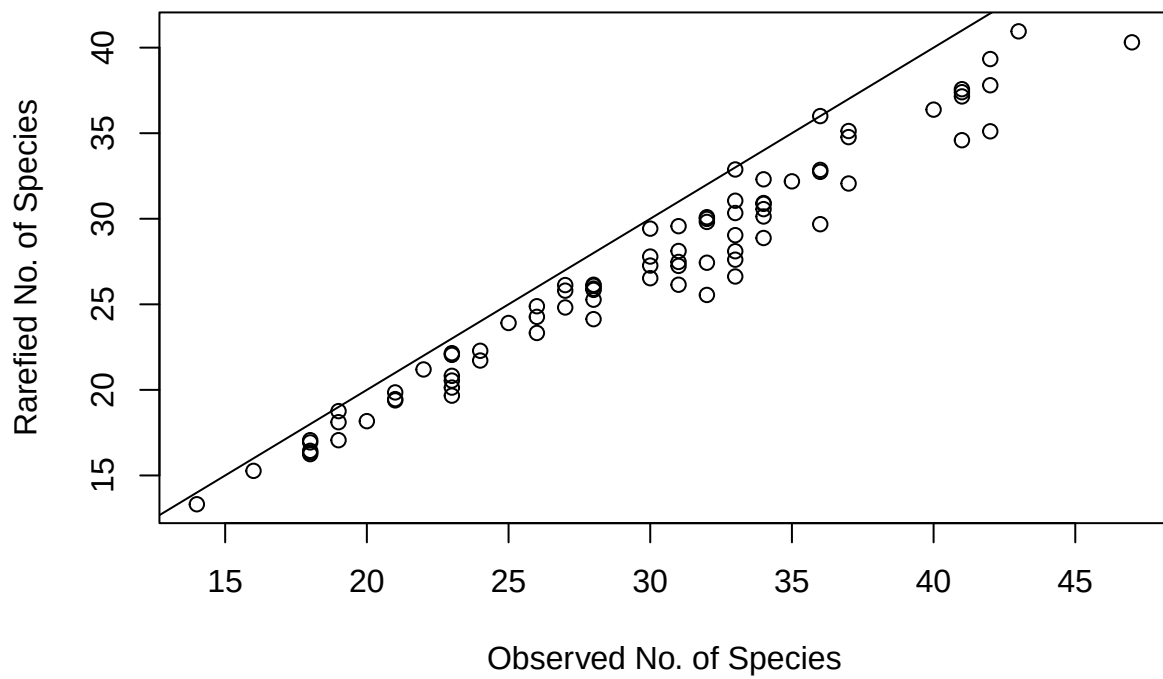
1. Richness with rarefaction

1.1. Palynological richness with rarefaction

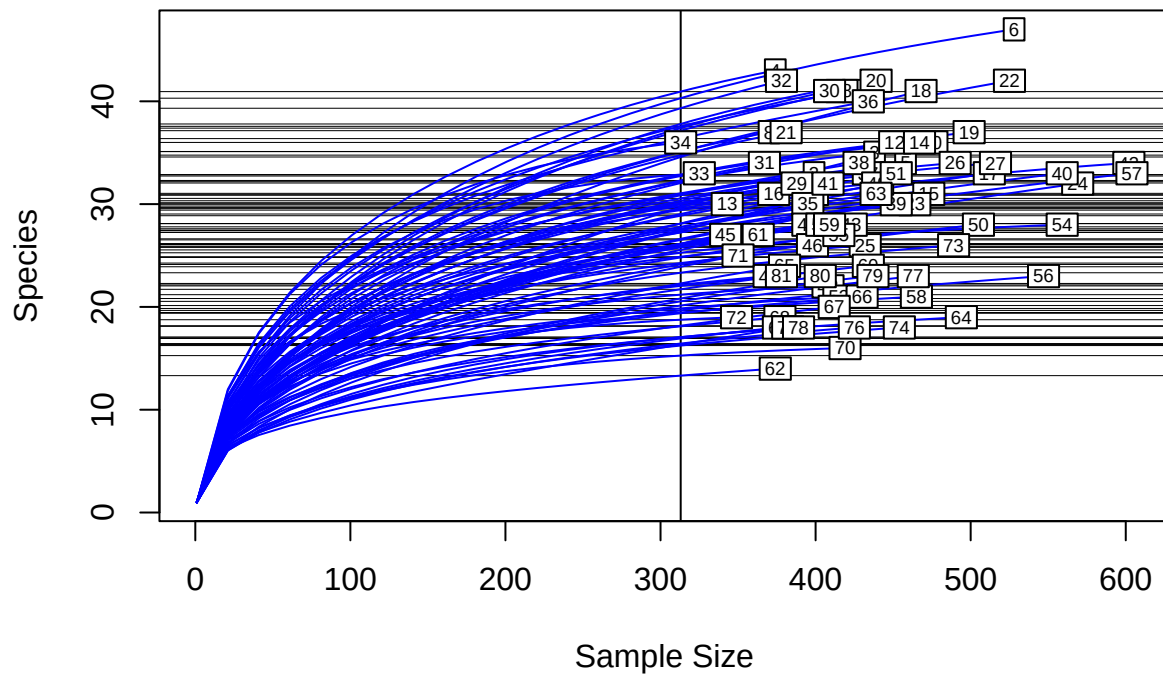
Function `rarefy` gives the expected richness in random subsamples of size “sample” from the community. The size of “sample” should be smaller than total community size. If “sample” is a vector, rarefaction of all observations is performed for each sample size separately

```
tram20_pollen_holocene_richness <- tram20_pollen_holocene %>%
  dplyr::select(-c(SampleId, age, depth, `Terrestrial pollen sum`:Lycopodium, `Evergreen trees`:Trees, S

S <- specnumber(tram20_pollen_holocene_richness) # observed number of species
raremax <- min(rowSums(tram20_pollen_holocene_richness))
Srare <- rarefy(tram20_pollen_holocene_richness, raremax)
plot(S, Srare, xlab = "Observed No. of Species", ylab = "Rarefied No. of Species")
abline(0, 1)
```



```
rarecurve(tram20_pollen_holocene_richness, step = 20, sample = raremax, col = "blue", cex = 0.6)
```



Create dataframe with rarefaction richness as a variable

```
tram20_pollen_holocene_rarefaction <- tram20_pollen_holocene %>%
  cbind(Srare) %>% #add rarefaction as a new column
  rename(Rarefaction = Srare) %>%
  relocate(Rarefaction, .after = "No of taxa")
```

2. Evenness

Create a function that allows for the estimation of different diversity indexes

```
calculate_evenness <- function(abundances, method = "pielou") {
  S <- specnumber(abundances)
  if (method == "pielou") {
    H <- diversity(abundances, "shannon")
    J <- H / log(S)
    return(J)
  } else if (method == "simpson") {
    lambda <- 1 - diversity(abundances, "simpson")
    N2 <- 1 / lambda
    E <- N2 / S
    return(E)
  }
}
```

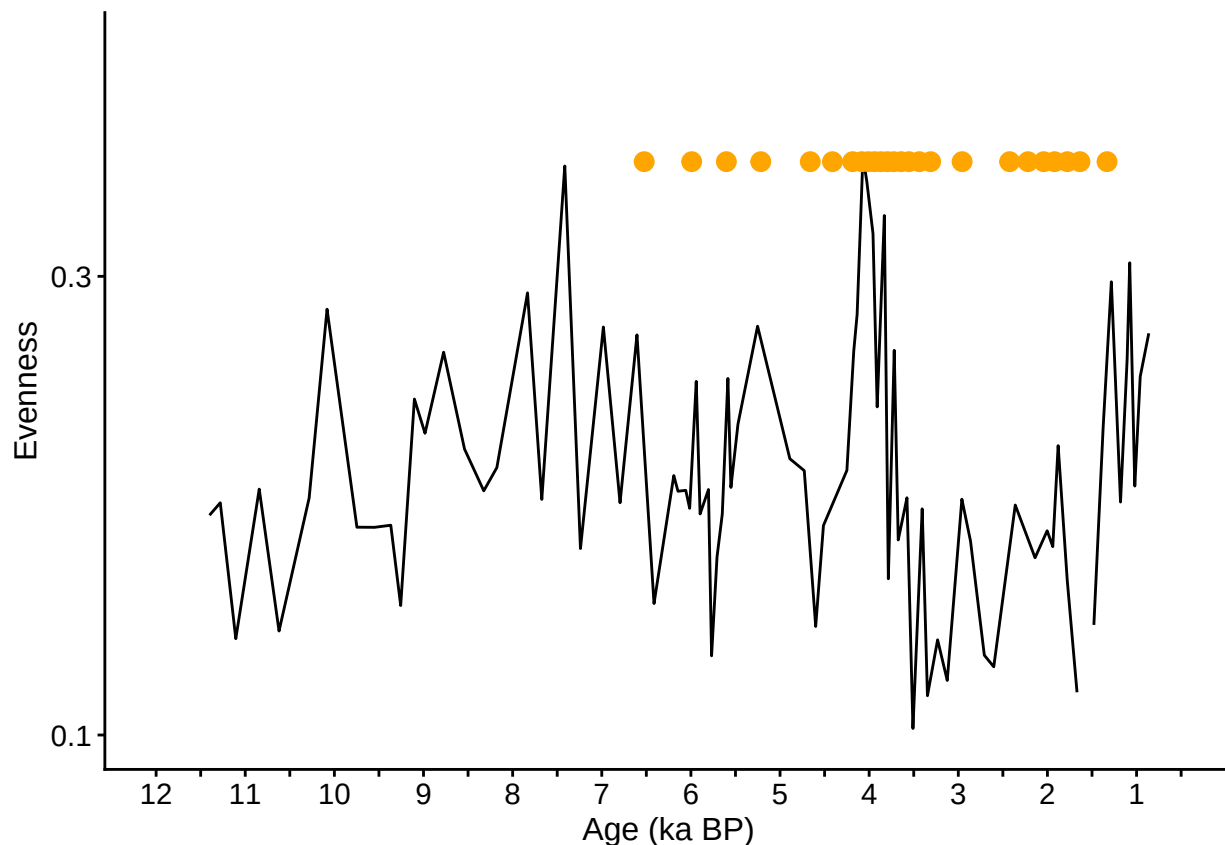
2.1. Palynological evenness

Calculate Simpson index and plot

```
evenness_simpson <- calculate_evenness(tram20_PAR %>%
  dplyr::select(-c(depth, age)), "simpson")

tram_pollen_evenness_simpson <- data.frame(tram20_PAR %>% dplyr::select(age),
  evenness_simpson)

tram_pollen_evenness_simpson %>%
  ggplot(aes(x = age/1000, y = evenness_simpson)) +
  geom_line() +
  scale_x_reverse(limits = c(12, 0.5),
    breaks = seq(0.5, 12, by = 0.5),
    labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
  scale_y_continuous(limits = c(0.1, 0.4),
    breaks = seq(0.1, 0.4, 0.2)) +
  theme_classic() +
  labs(x = "Age (ka BP)",
    y = "Evenness") +
  geom_point(data = RAI_animals_domestic %>%
    mutate(presence = ifelse(Cattle|Sheep|Goat > 0.0, 0.35, 0)), # orange points when any do
    aes(x = age/1000, y = presence), color = "orange", size = 3) +
  theme(axis.title.x = element_text(size = 12),
    axis.title.y = element_text(size = 12),
    axis.text.x = element_text(size = 11),
    axis.text.y = element_text(size = 11))
```



2.2. sedaDNA evenness

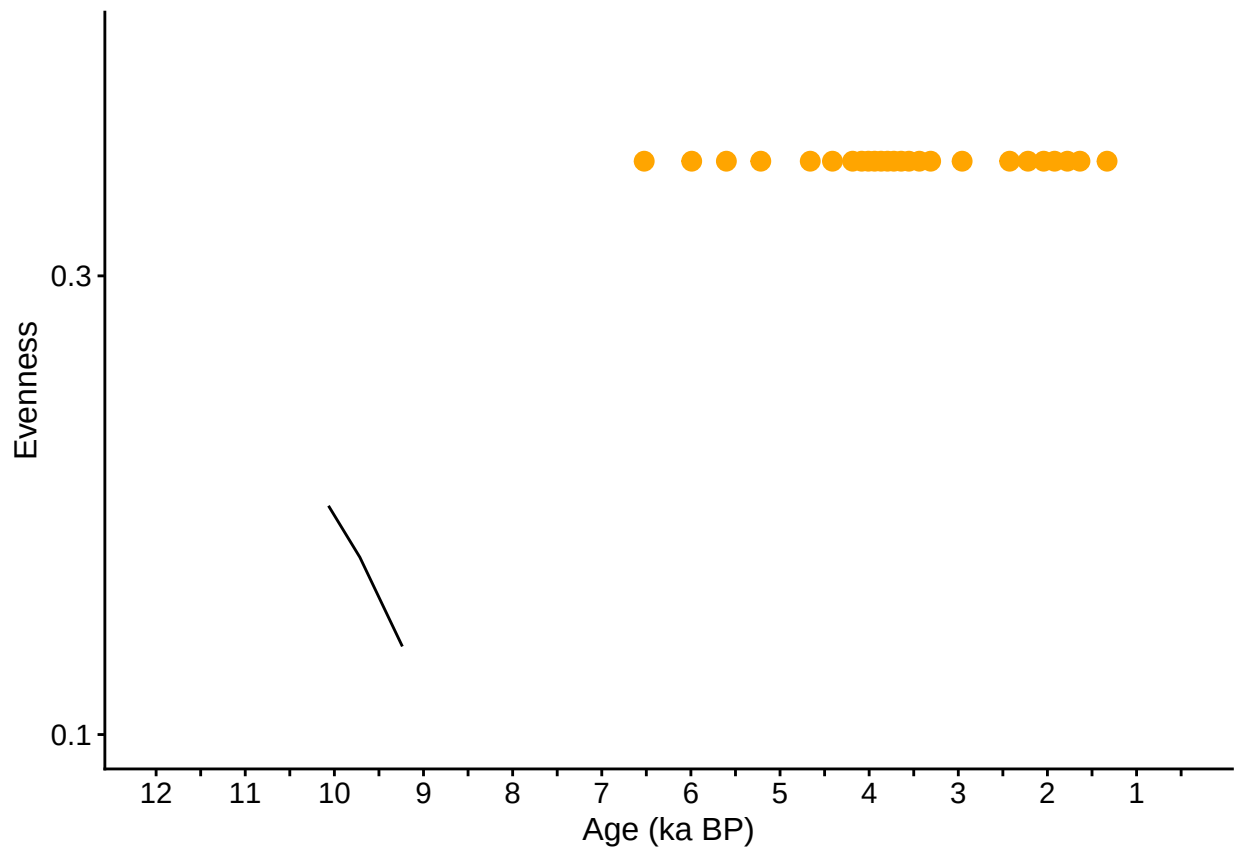
Calculate Simpson index and plot

```
evenness_simpson_RAI <- calculate_evenness(RAI_plants %>%
  dplyr::select(-c(sample_id, depth, age)), "simpson")

tram_RAI_evenness_simpson <- data.frame(RAI_plants %>% dplyr::select(age),
  evenness_simpson_RAI)

tram_RAI_evenness_simpson %>%
  ggplot(aes(x = age/1000, y = evenness_simpson_RAI)) +
  geom_line() +
  scale_x_reverse(limits = c(12, 0.5),
    breaks = seq(0.5, 12, by = 0.5),
    labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
  scale_y_continuous(limits = c(0.1, 0.4),
    breaks = seq(0.1, 0.4, 0.2)) +
  theme_classic() +
  labs(x = "Age (ka BP)",
    y = "Evenness") +
  geom_point(data = RAI_animals_domestic %>%
    mutate(presence = ifelse(Cattle|Sheep|Goat > 0.0, 0.35, 0)), # orange points when any do
    aes(x = age/1000, y = presence), color = "orange", size = 3) +
  theme(axis.title.x = element_text(size = 12),
```

```
axis.title.y = element_text(size = 12),
axis.text.x = element_text(size = 11),
axis.text.y = element_text(size = 11))
```



3. Rate of Change

3.1. Rate of Change of PAR

3.1.1. Import data for Rate of Change

```
tram20_RoC_PAR <- tram20_PAR %>%
  left_join(tram20_depth_ages_sed_rate %>%
    dplyr::select(SampleId, age), by = "age") %>%
  relocate(SampleId, .before = depth) %>%
  dplyr::select(-c(depth, age, Cyperaceae:Undetermined)) %>% #keep only terrestrial pollen types
  rename("sample_id" = "SampleId")

tram20_RoC_PAR %>%
  mutate(sum = rowSums(across(where(is.numeric)))) %>%
  dplyr::select(sample_id, sum) %>%
  filter(sum == min(sum)) #307
```

```
##           sample_id      sum
## 1 TRAM20-1B-4U-25-26 307.3769
```

3.1.2. BChron age depth model

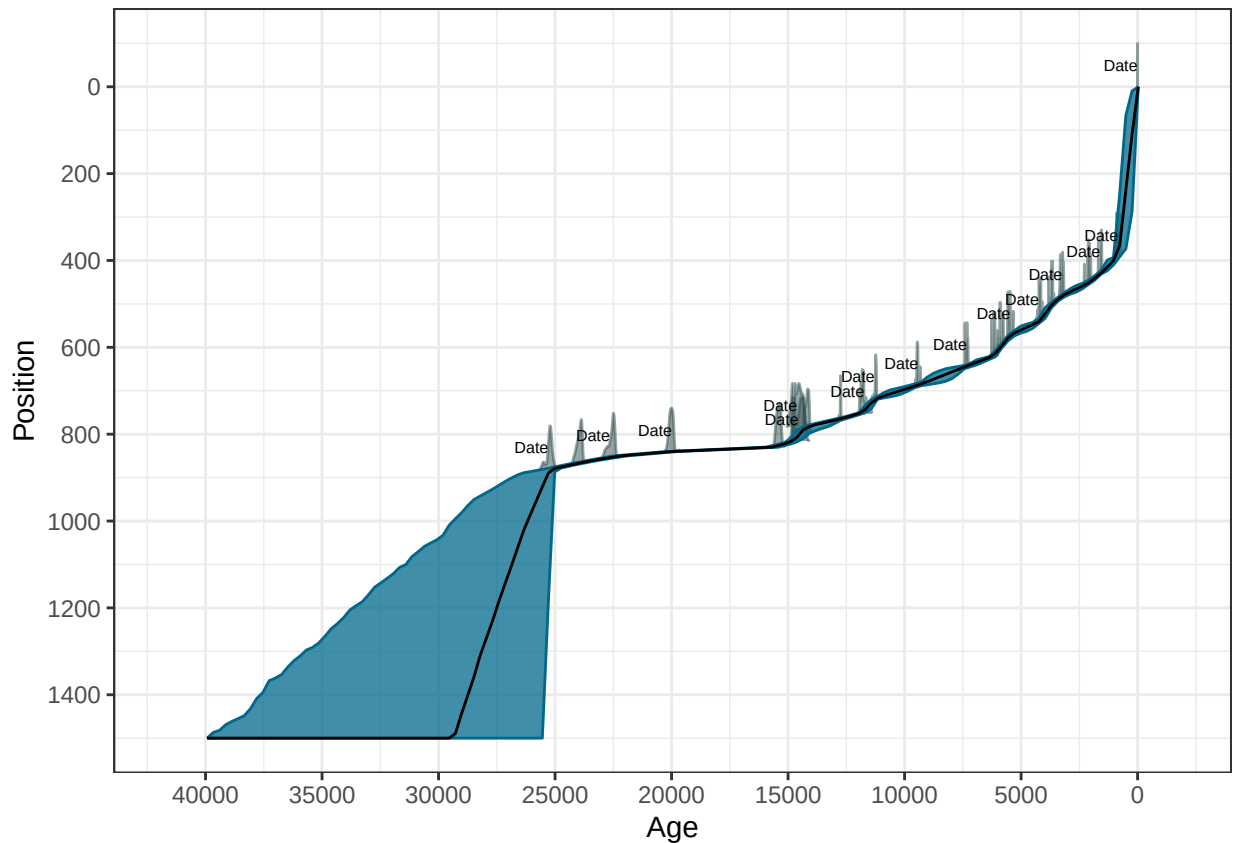
```
head(tram20_BChron_data)
```

```
##           id ages ageSds position thickness calCurves
## 1           Top-1    0      1      0.0          1   normal
## 2 TRAM20-1B-3U 19-20  920    23   390.9          1 intcal20
## 3 TRAM20-1B-3U 42-44 1687    21   429.5          1 intcal20
## 4 TRAM20-1B-3U 56-58 2126    23   452.0          1 intcal20
## 5 TRAM20-1B-3U 75-76 3033    27   480.9          1 intcal20
## 6 TRAM20-1B-3U 88-89 3435    23   501.8          1 intcal20
```

```
Tram20Chron = with(tram20_BChron_data, Bchronology(ages=ages,
                                                    ageSds=ageSds,
                                                    calCurves=calCurves,
                                                    positions=position,
                                                    positionThicknesses=thickness,
                                                    ids=id,
                                                    predictPositions=seq(0,1500,by=10)))
```

```
## |
```

```
plot(Tram20Chron)
```



Age uncertainties from BChron age-depth model

```
head(tram20_level) # file with sample_id and depths associated
```

```
## # A tibble: 6 x 2
##   sample_id      depth
##   <chr>         <dbl>
## 1 TRAM20-1B-3U-19-20 390
## 2 TRAM20-1B-3U-22-23 395
## 3 TRAM20-1B-3U-24-25 398
## 4 TRAM20-1B-3U-27-28 403
## 5 TRAM20-1B-3U-29-30 406
## 6 TRAM20-1B-3U-32-33 411
```

```
age_position <- Bchron::predict.BchronologyRun(object = Tram20Chron, newPositions = tram20_level$depth)
```

```
## |
```

```
age_uncertainties <- age_position %>% # assign ages to each depth
  as.data.frame() %>%
  mutate_all(., as.integer) %>%
  as.matrix()
colnames(age_uncertainties) <- tram20_level$sample_id
```



```
tram20_level_predicted <- tram20_level %>%
  mutate(age = apply(age_uncertainties, 2, stats::quantile, probs = 0.5))

head(tram20_level_predicted)
```

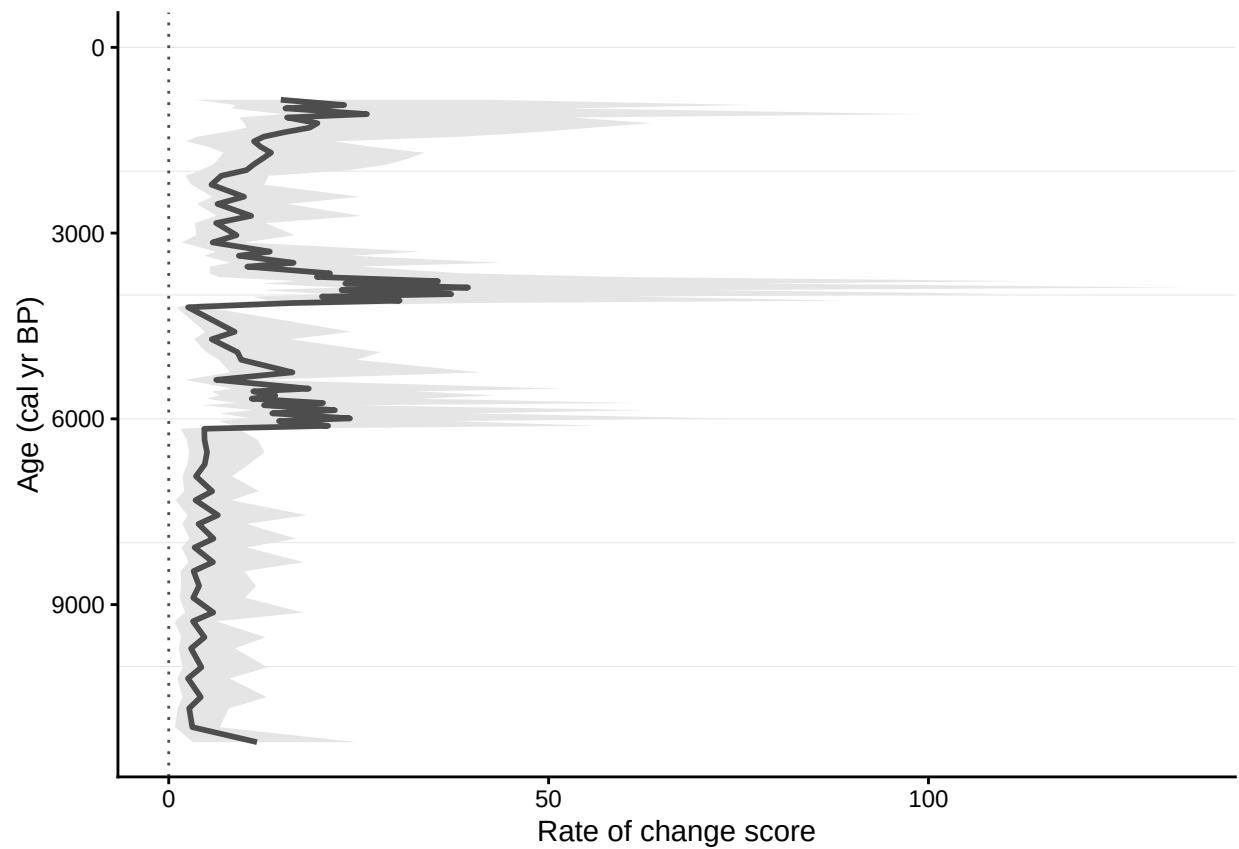
```
## # A tibble: 6 x 3
##   sample_id      depth  age
##   <chr>        <dbl> <dbl>
## 1 TRAM20-1B-3U-19-20    390  845
## 2 TRAM20-1B-3U-22-23    395  930.
## 3 TRAM20-1B-3U-24-25    398  985
## 4 TRAM20-1B-3U-27-28    403 1080
## 5 TRAM20-1B-3U-29-30    406 1135
## 6 TRAM20-1B-3U-32-33    411 1226
```

```
age.RoC.PAR <- tram20_depth_ages %>%
  rename("age" = "median") %>%
  dplyr::select(SampleId, age) %>%
  mutate(age = as.numeric(age)) %>%
  rename("sample_id" = "SampleId") %>%
  filter(sample_id %in% tram20_RoC_PAR$sample_id)
```

3.1.3. Perform Rate of Change

Scenario 4 - Estimating RoC for each level subsampling data and calculating age uncertainties

```
set_randomisations <- 1000
scenario_4_PAR <- RRatepol::estimate_roc(data_source_community = tram20_RoC_PAR,
                                         data_source_age = age.RoC.PAR,
                                         dissimilarity_coefficient = "chisq",
                                         working_units = "levels",
                                         time_standardisation = 500,
                                         smooth_method = "shep",
                                         standardise = TRUE,
                                         n_individuals = 29, # set the number of PAR
                                         rand = set_randomisations,
                                         use_parallel = TRUE,
                                         age_uncertainty = age_uncertainties) # Add the uncertainty
RRatepol::plot_roc(data_source = scenario_4_PAR)
```

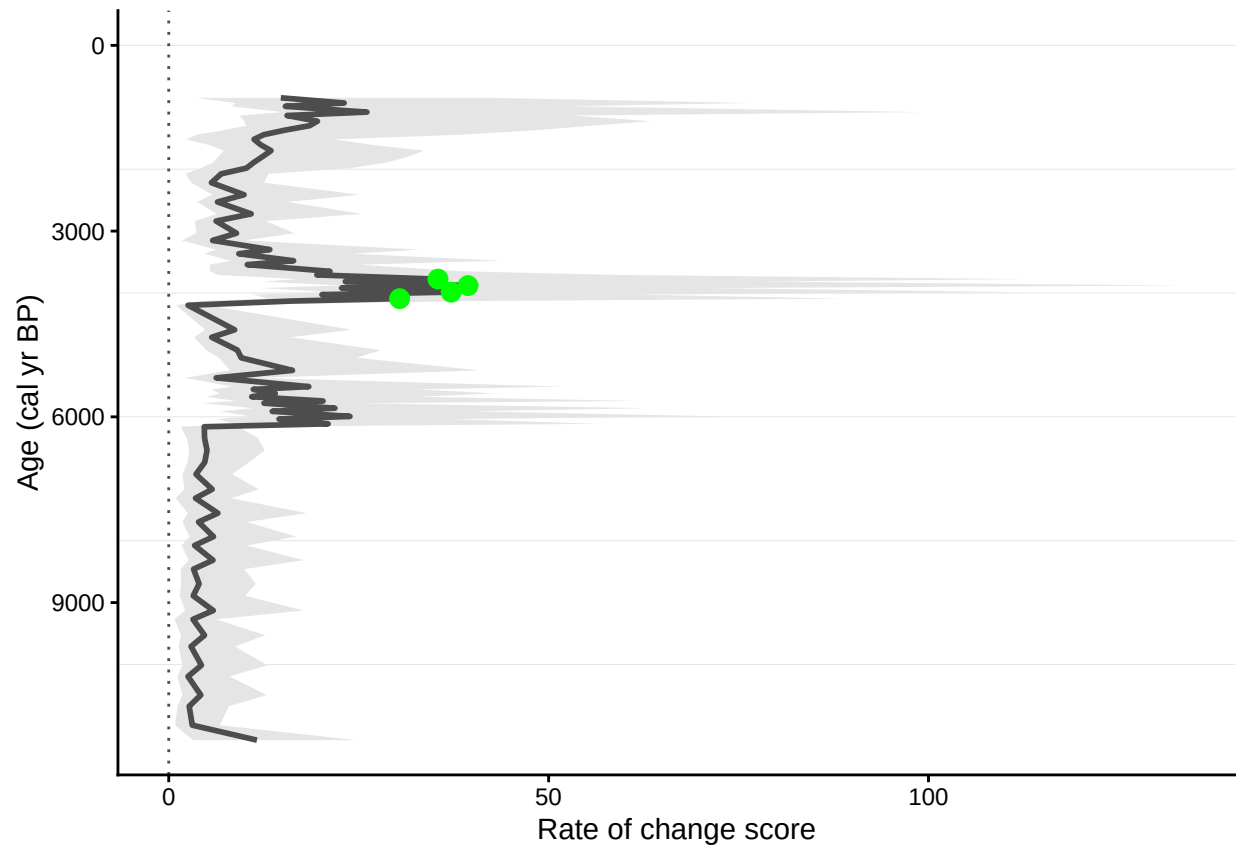


Scenario 4b - Estimating RoC detecting peak points

```
scenario_4_PAR_peak <- RRatepol::detect_peak_points(data_source = scenario_4_PAR,
                                                    sel_method = "trend_non_linear")
scenario_4_PAR_peak$Peak <- as.vector(scenario_4_PAR_peak$Peak)

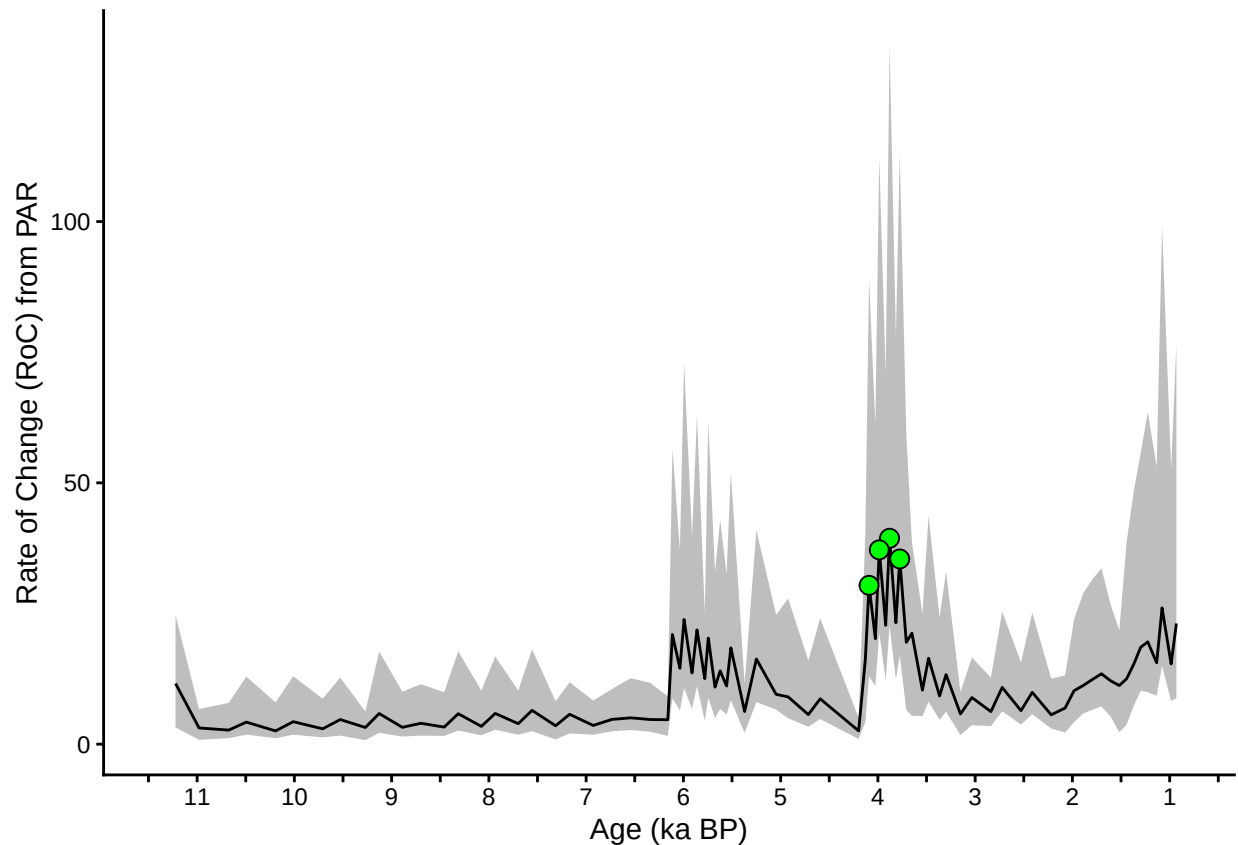
scenario_4_PAR_peak <- scenario_4_PAR_peak %>%
  mutate(ROC_peak = case_when (Peak == "TRUE" ~ ROC,
                              Peak == "FALSE" ~ NA))

RRatepol::plot_roc(data_source = scenario_4_PAR_peak, peaks = TRUE)
```



Plot

```
scenario_4_PAR_peak %>%
  ggplot(aes(x = Age/1000)) +
  theme_classic() +
  geom_ribbon(aes(ymin = ROC_dw, ymax = ROC_up), fill = "grey") +
  geom_line(aes(y = ROC)) +
  geom_point(size = 3, aes(y = ROC_peak), shape = 21, fill = "green") +
  scale_x_reverse(limits = c(max(CHAR$age)/1000, min(CHAR$age)/1000),
                  breaks = seq(0.5, 12, by = 0.5),
                  labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
                                  "")) +
  xlab("Age (ka BP)") +
  ylab("Rate of Change (RoC) from PAR")
```



3.2. Rate of Change with SAR of coprophilous

3.2.1. Import data for Rate of Change

```
tram20_RoC_SAR <- tram20_SAR_coprophilous %>%
  dplyr::select(-age) %>% #exclude age
  left_join(tram20_level, by = "depth") %>%
  dplyr::select(-depth) %>%
  relocate(sample_id, .before = 0)

tram20_RoC_SAR %>%
  mutate(sum = rowSums(across(where(is.numeric)))) %>%
  dplyr::select(sample_id, sum) %>%
  filter(sum > 0) %>%
  filter(sum == min(sum)) #2.57
```

```
## # A tibble: 1 x 2
##   sample_id      sum
##   <chr>         <dbl>
## 1 TRAM20-1B-4U-103-104 2.57
```

```
age.RoC.SAR <- age.RoC.PAR %>%
  filter(age %in% tram20_SAR$age) %>%
```

```

dplyr::select(sample_id, age)

age_uncertainties_SAR <- age_uncertainties %>%
  as.data.frame() %>%
  dplyr::select(any_of(age.RoC.SAR$sample_id)) %>%
  as.matrix()

```

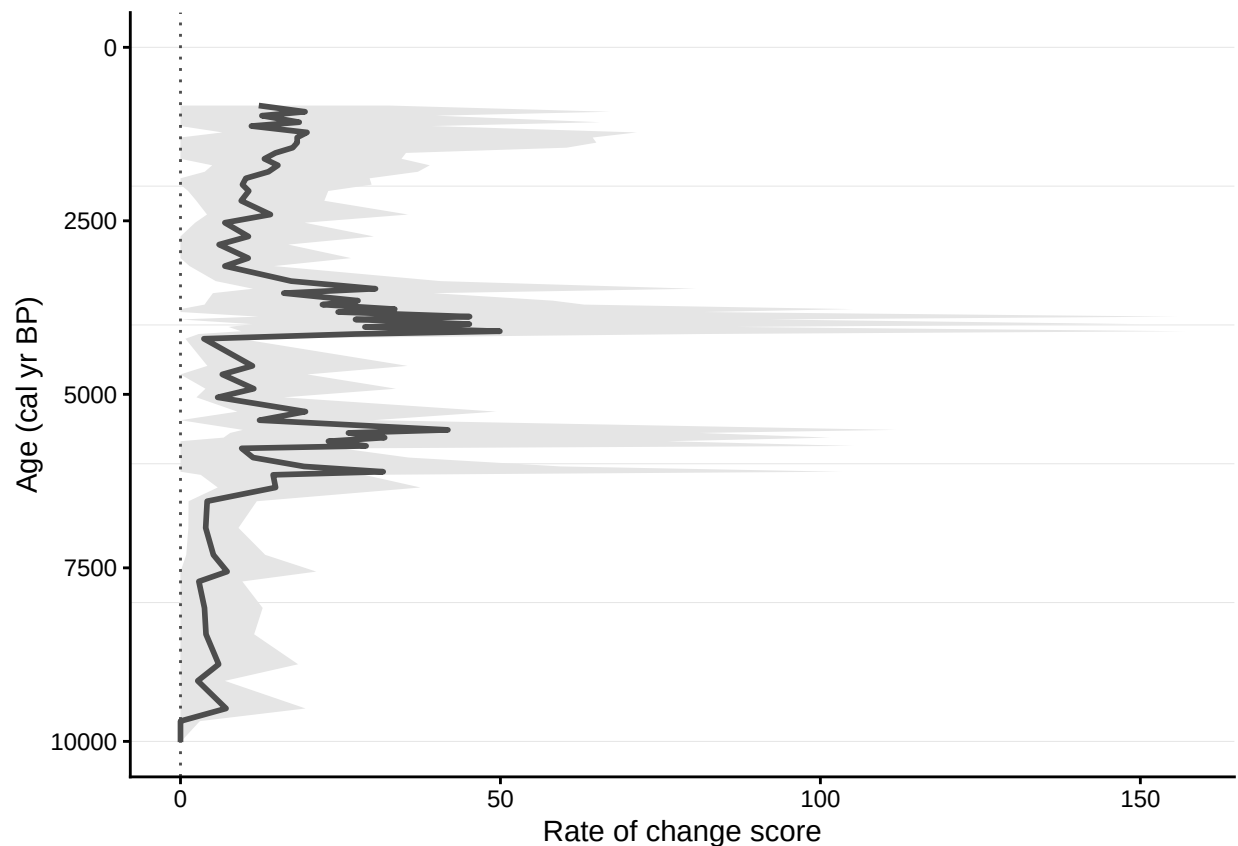
3.2.2. Perform Rate of Change

Scenario 4 - Estimating RoC for each level subsampling data and calculating age uncertainties

```

set_randomisations <- 1000
scenario_4_SAR <- RRatepol::estimate_roc(data_source_community = tram20_RoC_SAR,
                                         data_source_age = age.RoC.SAR,
                                         dissimilarity_coefficient = "chisq",
                                         working_units = "levels",
                                         time_standardisation = 500,
                                         smooth_method = "shep",
                                         standardise = TRUE,
                                         n_individuals = 3, # set the number of SAR
                                         rand = set_randomisations,
                                         use_parallel = TRUE,
                                         age_uncertainty = age_uncertainties_SAR) # Add the uncertainty
RRatepol::plot_roc(data_source = scenario_4_SAR)

```



Scenario 4b - Estimating RoC detecting peak points

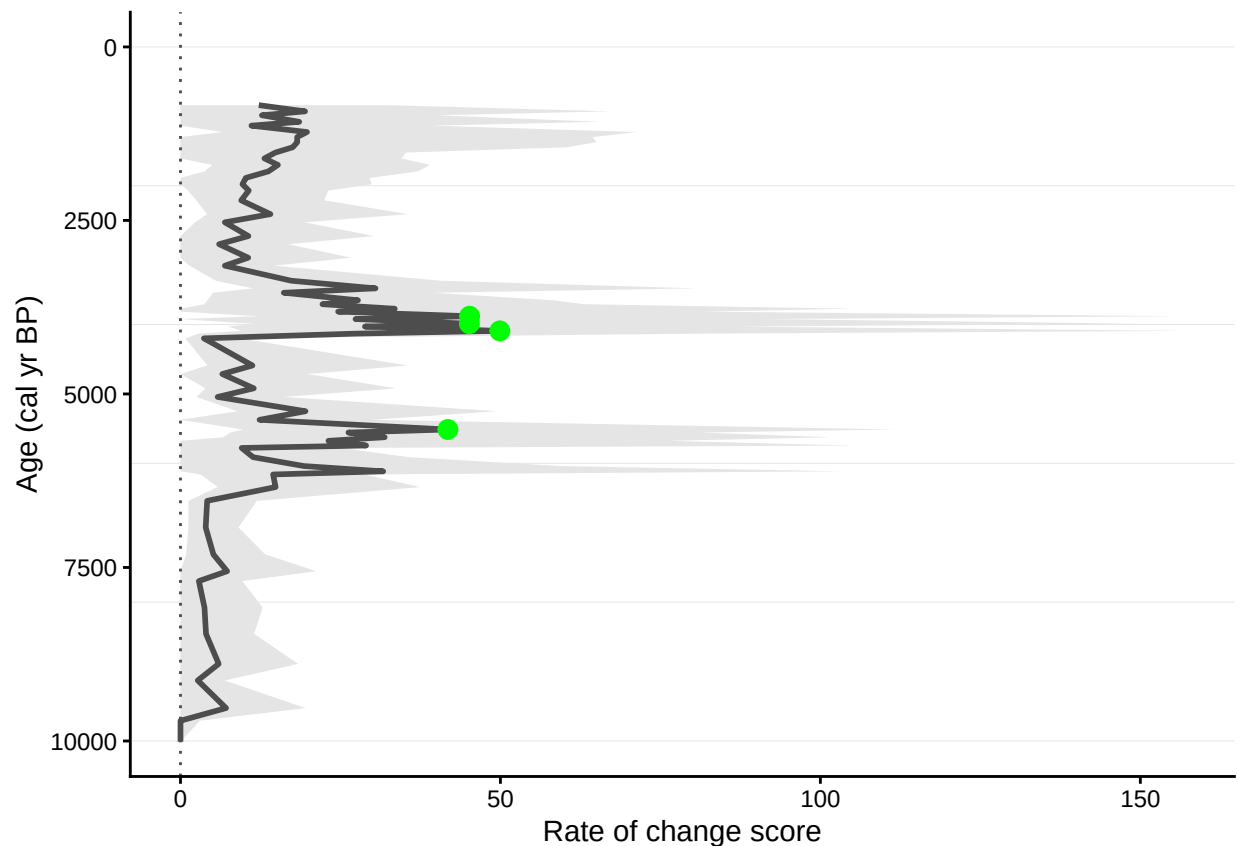
```
# first, solve problems associated to the presence of 0 in the dataframe
scenario_4_SAR_clean <- scenario_4_SAR %>%
  mutate(across(where(is.numeric), ~ replace(., . == 0, 0.00001)))

scenario_4_SAR_peak <- RRatepol::detect_peak_points(data_source = scenario_4_SAR_clean,
  sel_method = "trend_non_linear")

scenario_4_SAR_peak$Peak <- as.vector(scenario_4_SAR_peak$Peak)

scenario_4_SAR_peak <- scenario_4_SAR_peak %>%
  mutate(ROC_peak = case_when (Peak == "TRUE" ~ ROC,
    Peak == "FALSE" ~ NA))

RRatepol::plot_roc(data_source = scenario_4_SAR_peak, peaks = TRUE)
```



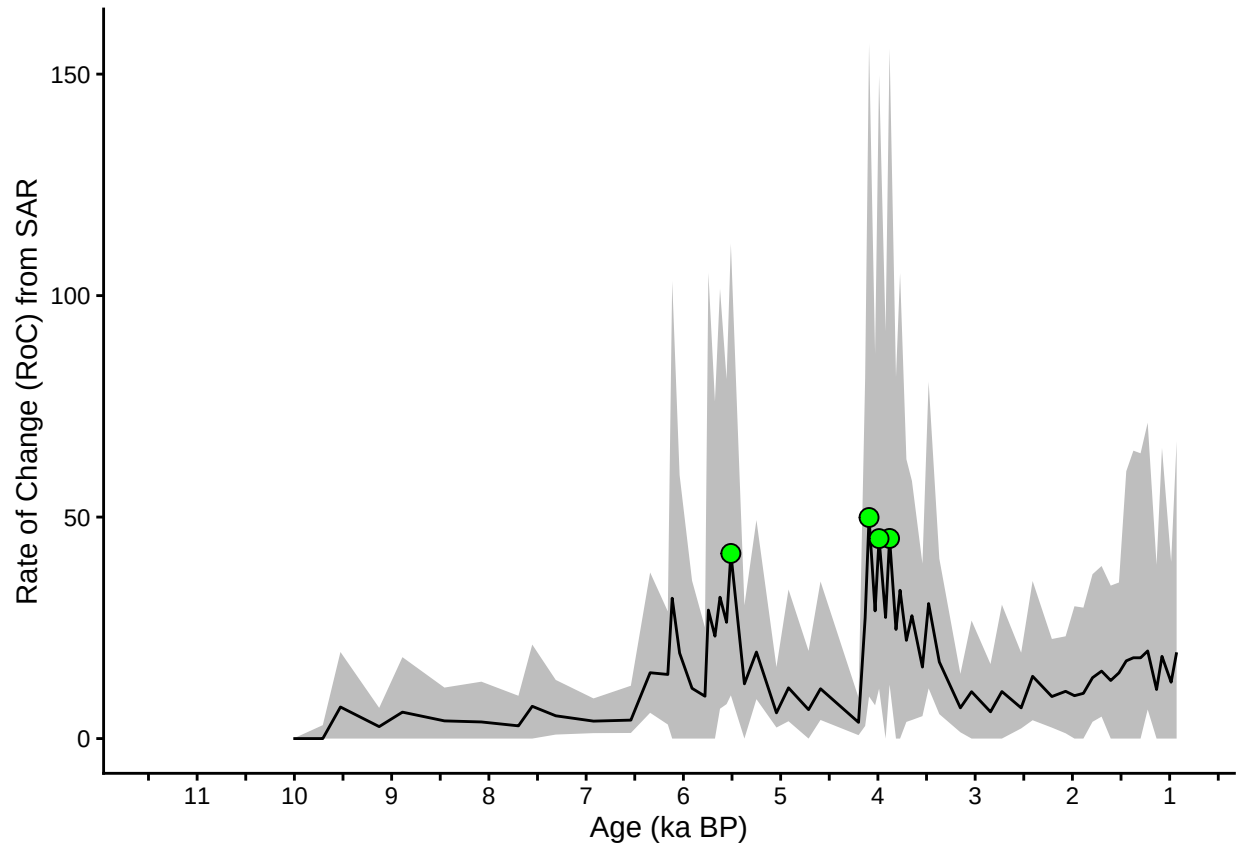
Plot

```
scenario_4_SAR_peak %>%
  ggplot(aes(x = Age/1000))+
  theme_classic() +
  geom_ribbon(aes(ymin = ROC_dw, ymax = ROC_up), fill = "grey") +
  geom_line(aes(y = ROC)) +
  geom_point(size = 3, aes(y = ROC_peak), shape = 21, fill = "green") +
  scale_x_reverse(limits = c(max(CHAR$age)/1000, min(CHAR$age)/1000),
```

```

breaks = seq(0.5, 12, by = 0.5),
labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
xlab("Age (ka BP)" ) +
ylab("Rate of Change (RoC) from SAR")

```



4. Change point analysis with CHAR data

```

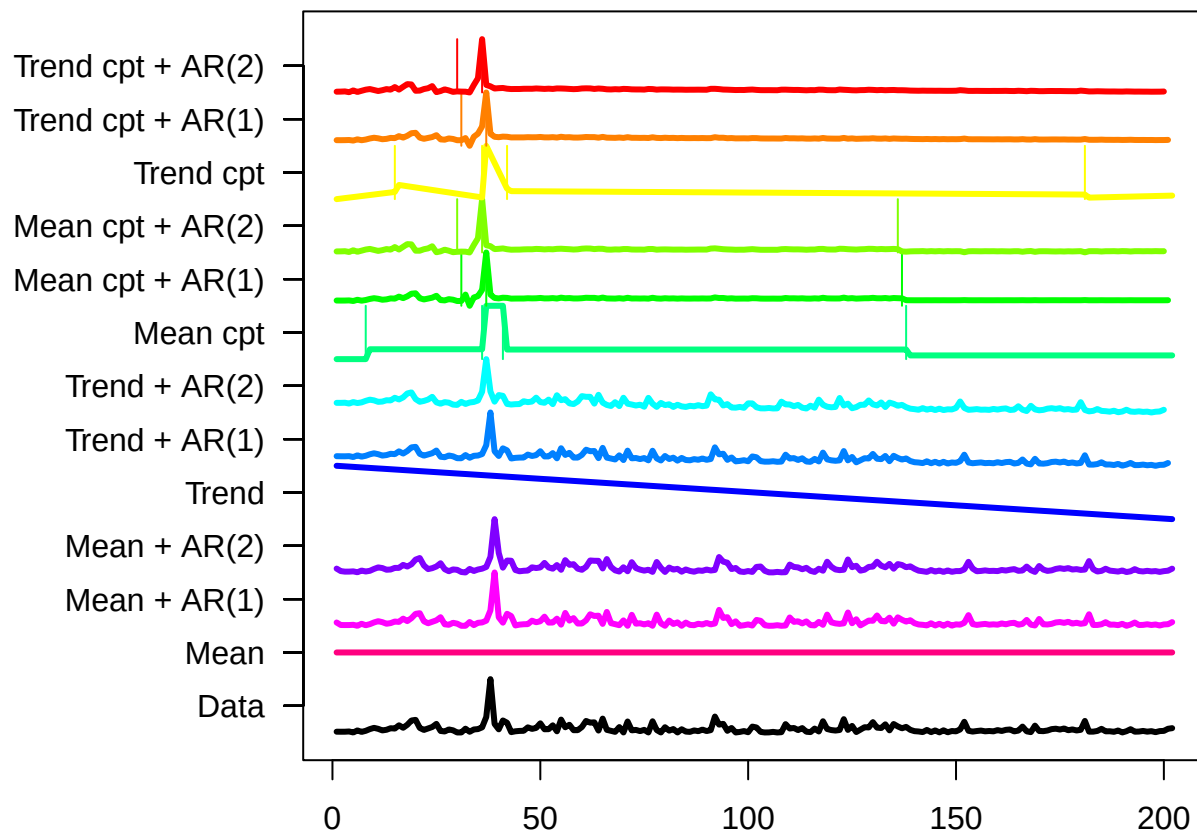
tram20_CHAR_cpt <- CHAR %>%
  dplyr::select(age, CHAR.tram) %>%
  arrange(age)

out=envcpt(tram20_CHAR_cpt$CHAR.tram) # <- The key instruction!

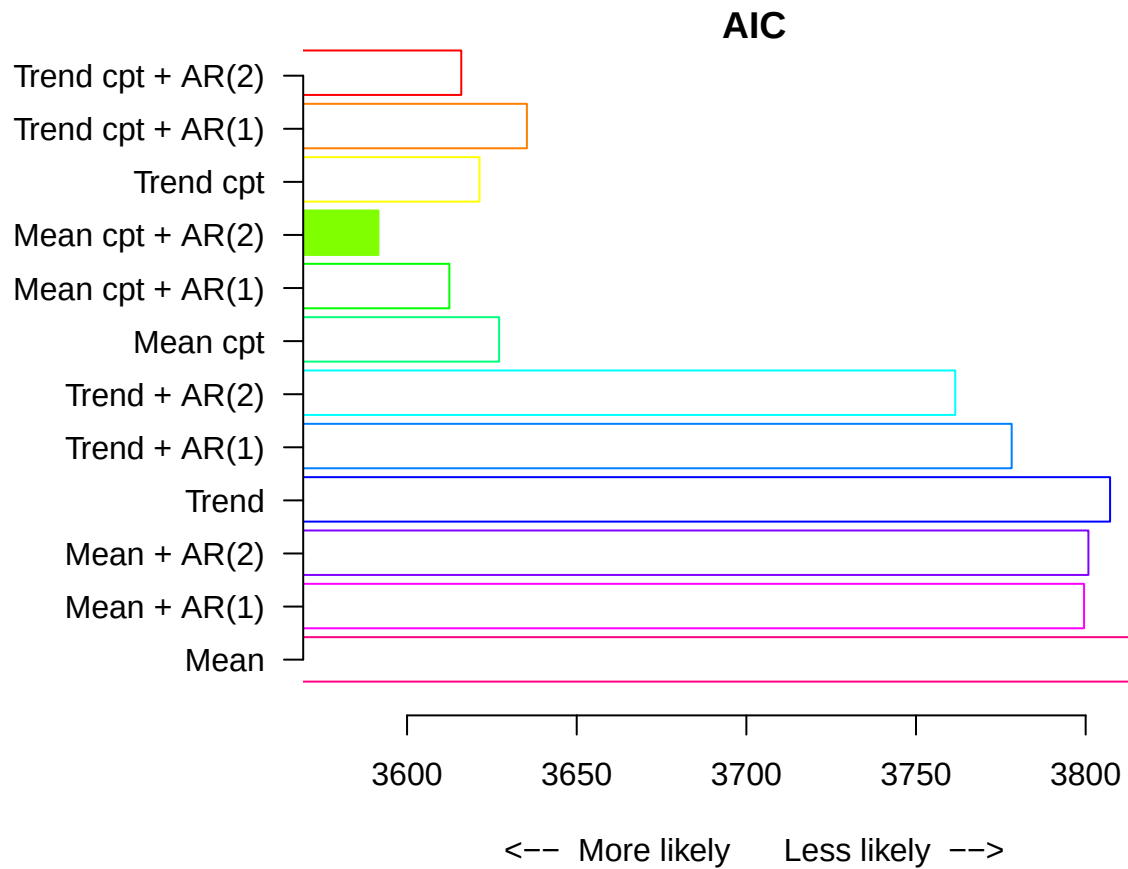
```

```
## |
```

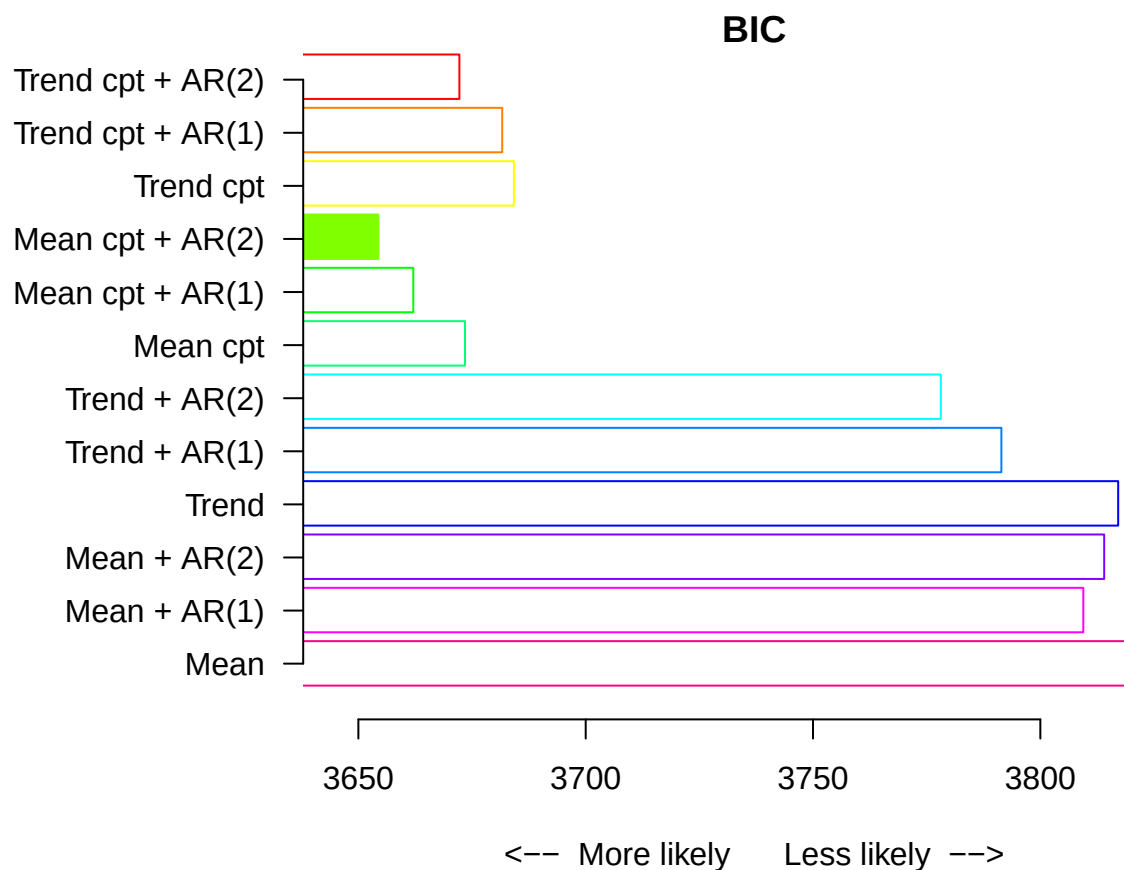
```
plot(out,type='fit') # plots the fits
```



```
plot(out,type="aic") # plots the aic values
```

```
plot(out,type="bic") # plots the bic values
```



```
which.min(AIC(out)); which.min(BIC(out))
```

```
## meanar2cpt
##          6
```

```
## meanar2cpt
##          6
```

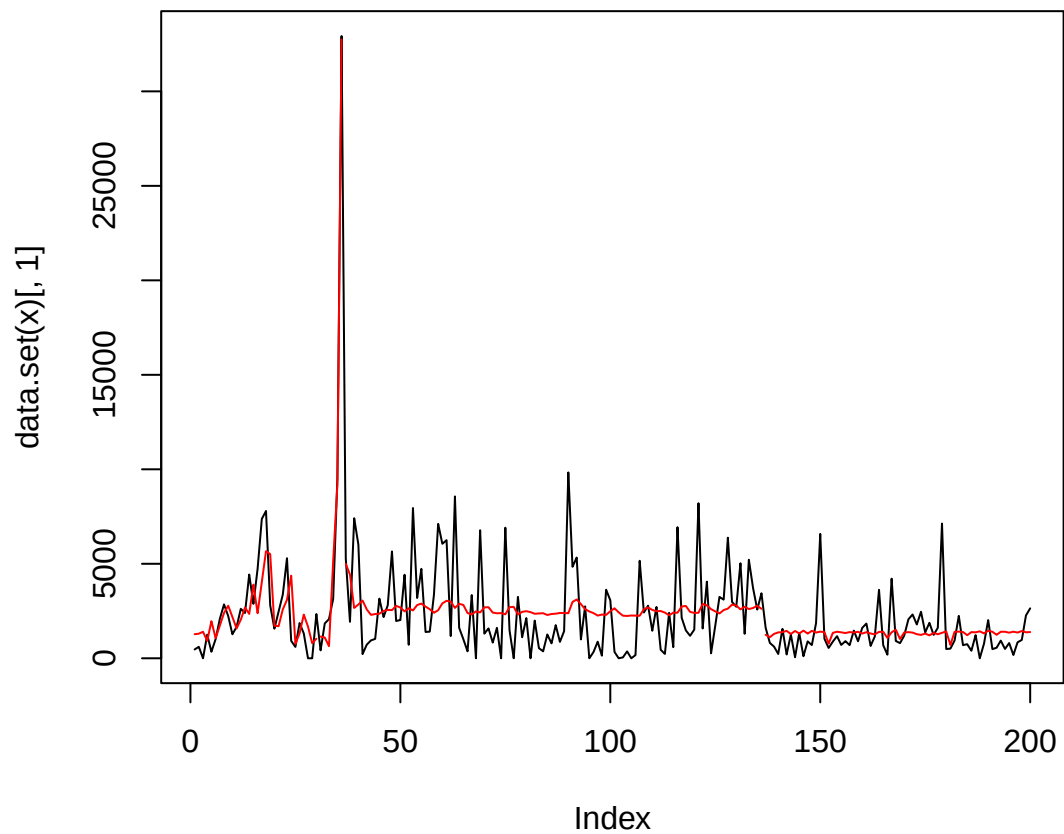
```
model_CHAR <- envcpt(tram20_CHAR_cpt$CHAR.tram, models = "meanar2cpt") #here the chosen model
```

```
## |
```

```
summary(model_CHAR$meanar2cpt)
```

```
## Created Using changepoint version 2.3
## Changepoint type      : Change in regression
## Method of analysis    : PELT
## Test Statistic       : Normal
## Type of penalty       : MBIC with value, 31.7899
## Maximum no. of cpts   : Inf
## Changepoint Locations : 30 36 136
```

```
plot(model_CHAR$meanar2cpt)
```



Assign ages to each changepoint and extract positions from changepoints

```
cpts_CHAR <- cpts(model_CHAR$meanar2cpt) # rows where changepoints are
print(cpts_CHAR)
```

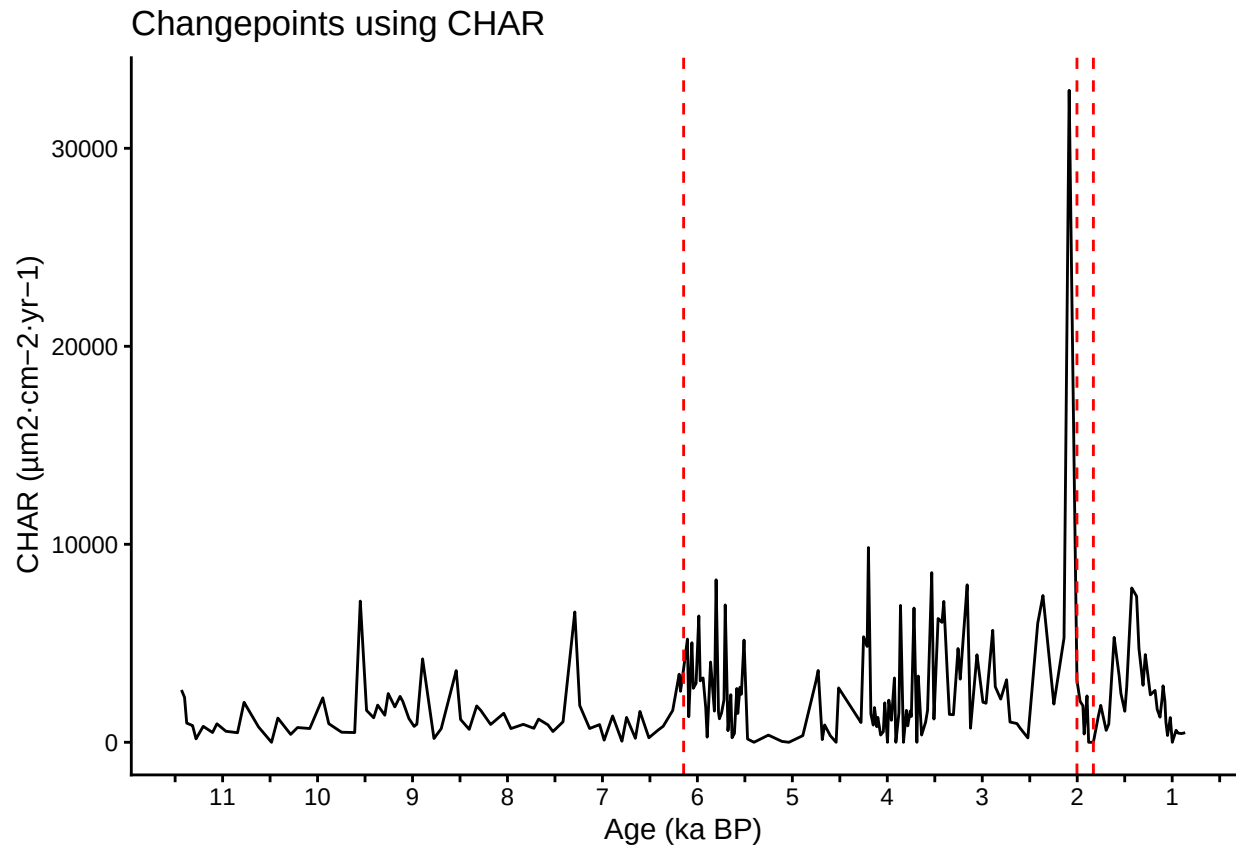
```
## [1] 30 36 136
```

```
coef_ar_CHAR <- model_CHAR$meanar2cpt@param.est$ar
```

```
changepoint_ages_CHAR <- tram20_CHAR_cpt$age[cpts_CHAR] # ages to each row: 1830 2003 6145
```

Plot

```
tram20_CHAR_cpt %>%
  ggplot(aes(x = age/1000, y = CHAR.tram)) +
  geom_line() +
  geom_vline(xintercept = changepoint_ages_CHAR/1000, color = "red", linetype = "dashed") +
  labs(x = "Age (ka BP)", y = "CHAR ( $\mu\text{m}^2\cdot\text{cm}^{-2}\cdot\text{yr}^{-1}$ )", title = "Changepoints using CHAR") +
  theme_classic() +
  scale_x_reverse(limits = c(max(CHAR$age)/1000, min(CHAR$age)/1000),
    breaks = seq(0.5, 12, by = 0.5),
    labels = ifelse(seq(0.5, 12, by = 0.5) %% 1 == 0, as.character(seq(0.5, 12, by = 0.5)),
```



5. Cross-correlations for different periods: before and after 6.5 ka BP

Import data

```
char_full <- CHAR %>%
  filter(age %in% tram20_pollen_holocene$age)
```

Create a function to perform cross-correlations for both periods

```
tram_ccf <- function(data, taxa, char_full_data, period = "all",
  age_threshold = 6500, lag_max = 5) {

  # Filter data based on period
  filtered_data <- switch(period,
    "all" = data,
    "before" = data %>% filter(age > age_threshold),
    "after" = data %>% filter(age <= age_threshold))

  # Filter CHAR data to match the same period
  char_filtered <- char_full_data %>%
    filter(age %in% filtered_data$age) %>%
    dplyr::select(Char.tram) %>%
```

```

pull()

# Transform taxa data into temporal series
taxa_ts <- filtered_data %>%
  dplyr::select({{taxa}}) %>%
  pull()

# Detrending lineal: adjust simple linear model and extract residuals
taxa_det <- resid(lm(taxa_ts ~ seq_along(taxa_ts)))
char_det <- resid(lm(char_filtered ~ seq_along(char_filtered)))

# Cross-correlation
ccf_result <- ccf(taxa_det, char_det,
  lag.max = lag_max,
  plot = FALSE)

ccf_df <- with(ccf_result, data.frame(lag, acf))

n_obs <- length(taxa_det) # Significance
ci <- qnorm((1 + 0.95) / 2) / sqrt(n_obs) # 95%

# Plot title with period info
period_title <- switch(period,
  "all" = "All Period",
  "before" = paste("age >", age_threshold),
  "after" = paste("age <=", age_threshold))

# Plot
if (period == "before") {
  p <- ggplot(ccf_df, aes(x = lag, y = acf)) +
    geom_bar(stat = "identity", width = 1,
      colour = "#666666", fill = "grey") +
    geom_hline(yintercept = c(-ci, ci),
      linetype = "dashed", color = "red") +
    labs(y = paste(deparse(substitute(taxa))),
      x = "Lags") +
    theme_classic() +
    geom_vline(xintercept = 0, linetype = "dashed", color = "#666666") +
    theme(axis.title.x = element_blank(),
      axis.title.y = element_text(face = "italic", size = 12, angle = 0, vjust = 0.5),
      axis.text.x = element_text(size = 11),
      axis.text.y = element_text(size = 11))
} else if (period == "after") {
  p <- ggplot(ccf_df, aes(x = lag, y = acf)) +
    geom_bar(stat = "identity", width = 1,
      colour = "#666666", fill = "grey") +
    geom_hline(yintercept = c(-ci, ci),
      linetype = "dashed", color = "red") +
    theme_classic() +
    geom_vline(xintercept = 0, linetype = "dashed", color = "#666666") +
    theme(axis.title.x = element_blank(),
      axis.title.y = element_blank(),

```

```

    axis.text.x = element_text(size = 11),
    axis.text.y = element_text(size = 11))
  }

  return(list(plot = p, ccf_df = ccf_df, ci = ci, n = n_obs, period = period))
}

```

5.1. Cross-correlations for each taxa and periods

Abies

```

abies_ccf_before <- tram_ccf(tram20_PAR, Abies, char_full, period = "before")
abies_ccf_after <- tram_ccf(tram20_PAR, Abies, char_full, period = "after")

```

Pinus

```

pinus_ccf_before <- tram_ccf(tram20_PAR, Pinus, char_full, period = "before")
pinus_ccf_after <- tram_ccf(tram20_PAR, Pinus, char_full, period = "after")

```

Fagus

```

fagus_ccf_before <- tram_ccf(tram20_PAR, Fagus, char_full, period = "before")
fagus_ccf_after <- tram_ccf(tram20_PAR, Fagus, char_full, period = "after")

```

Betula

```

betula_ccf_before <- tram_ccf(tram20_PAR, Betula, char_full, period = "before")
betula_ccf_after <- tram_ccf(tram20_PAR, Betula, char_full, period = "after")

```

Corylus

```

corylus_ccf_before <- tram_ccf(tram20_PAR, Corylus, char_full, period = "before")
corylus_ccf_after <- tram_ccf(tram20_PAR, Corylus, char_full, period = "after")

```

Deciduous Quercus

```

decquercus_ccf_before <- tram_ccf(tram20_PAR %>%
  rename("Dec. Quercus" = "Deciduous Quercus"),
  `Dec. Quercus`, char_full, period = "before")
decquercus_ccf_after <- tram_ccf(tram20_PAR %>%
  rename("Dec. Quercus" = "Deciduous Quercus"),
  `Dec. Quercus`, char_full, period = "after")

```

Poaceae

```

poaceae_ccf_before <- tram_ccf(tram20_PAR, Poaceae, char_full, period = "before")
poaceae_ccf_after <- tram_ccf(tram20_PAR, Poaceae, char_full, period = "after")

```

Palynological richness

```
richness_ccf_before <- tram_ccf(tram20_pollen_holocene_rarefaction %>%
  rename("Richness" = "Rarefaction"),
  Richness, char_full, period = "before")
richness_ccf_after <- tram_ccf(tram20_pollen_holocene_rarefaction %>%
  rename("Richness" = "Rarefaction"),
  Richness, char_full, period = "after")
```

Palynological evenness

```
evenness_ccf_before <- tram_ccf(tram_pollen_evenness_simpson%>%
  rename("Evenness" = "evenness_simpson"),
  Evenness, char_full, period = "before")
evenness_ccf_after <- tram_ccf(tram_pollen_evenness_simpson%>%
  rename("Evenness" = "evenness_simpson"),
  Evenness, char_full, period = "after")
```

5.2. Create plots of cross-correlations

```
p.abies.ccf_before <- abies_ccf_before$plot
p.abies.ccf_after <- abies_ccf_after$plot

p.pinus.ccf_before <- pinus_ccf_before$plot
p.pinus.ccf_after <- pinus_ccf_after$plot

p.fagus.ccf_before <- fagus_ccf_before$plot
p.fagus.ccf_after <- fagus_ccf_after$plot

p.betula.ccf_before <- betula_ccf_before$plot
p.betula.ccf_after <- betula_ccf_after$plot

p.corylus.ccf_before <- corylus_ccf_before$plot
p.corylus.ccf_after <- corylus_ccf_after$plot

p.decquercus.ccf_before <- decquercus_ccf_before$plot
p.decquercus.ccf_after <- decquercus_ccf_after$plot

p.poaceae.ccf_before <- poaceae_ccf_before$plot
p.poaceae.ccf_after <- poaceae_ccf_after$plot

p.richness.ccf_before <- richness_ccf_before$plot
p.richness.ccf_after <- richness_ccf_after$plot

p.evenness.ccf_before <- evenness_ccf_before$plot
p.evenness.ccf_after <- evenness_ccf_after$plot
```

6. Structural Equation Modelling (SEM)

Prepare all data

```
RAI_plants_richness <- RAI_plants %>%
  mutate(richness = rowSums(across(c(`Abies alba`, Ziziphora), ~ .x > 0), na.rm = TRUE))

RAI_plants_richness_binned <- RAI_plants_richness %>%
  mutate(binned_age = sapply(age, function(x) bins_pollen[which.min(abs(bins_pollen-x))])) %>%
  mutate(RAI_Fagus = Fagus + `Fagus sylvatica`) %>%
  group_by(binned_age) %>%
  summarise(richness = mean(richness, na.rm = TRUE),
            RAI_Fagus = mean(RAI_Fagus, na.rm = TRUE)) %>%
  rename("age" = "binned_age",
         "plant_richness_dna" = "richness") %>%
  ungroup()

head(RAI_plants_groups)
```

```
## # A tibble: 6 x 4
##   age `Abies alba` `Open landscape` `Novel herbaceous taxa`
##   <dbl>         <dbl>         <dbl>         <dbl>
## 1  1331         0.0142         0.0137         0.00679
## 2  1496         0.0909         0.167          0.0540
## 3  1635         0.0389         0.133          0.0643
## 4  1776         0.0334         0.0601         0.0813
## 5  1920         0.0721         0.0566         0.0551
## 6  2042         0.0634         0.0659         0.0792
```

```
RAI_plants_groups_binned <- RAI_plants_groups %>%
  mutate(binned_age = sapply(age, function(x) bins_pollen[which.min(abs(bins_pollen-x))])) %>%
  group_by(binned_age) %>%
  summarise(across(c(`Abies alba`, `Open landscape`, `Novel herbaceous taxa`),
                    ~ mean(., na.rm = TRUE))) %>%
  rename("age" = "binned_age",
         "RAI_Abies" = "Abies alba",
         "RAI_Open" = "Open landscape",
         "RAI_Novel" = "Novel herbaceous taxa") %>%
  ungroup()

tram_RAI_evenness_simpson_binned <- tram_RAI_evenness_simpson %>%
  mutate(binned_age = sapply(age, function(x) bins_pollen[which.min(abs(bins_pollen-x))])) %>%
  group_by(binned_age) %>%
  summarise(evenness_simpson_RAI = mean(evenness_simpson_RAI, na.rm = TRUE)) %>%
  rename("age" = "binned_age") %>%
  ungroup()

head(Mendukilo)
```

```
##   age depth..mm. d13C.Mendukilo
## 1   0         0        -8120
## 2  10       1000       -9500
## 3  10       2000       -9490
## 4  20       3000       -9030
## 5  20       4000       -8460
## 6  30       5000       -8440
```



```
Mendukilo_binned <- Mendukilo %>%
  mutate(binned_age = sapply(age, function(x) bins_pollen[which.min(abs(bins_pollen-x))])) %>%
  group_by(binned_age) %>%
  summarise(d13C.Mendukilo = mean(d13C.Mendukilo, na.rm = TRUE)) %>%
  rename("age" = "binned_age") %>%
  ungroup() %>%
  as.data.frame()
```

All data in one dataframe

```
data_SEM <- RAI_animals_domestic_binned %>%
  left_join(RAI_plants_richness_binned, by = "age") %>%
  left_join(Char %>%
    dplyr::select(Char.tram, age), by = "age") %>%
  left_join(tram_pollen_evenness_simpson, by = "age") %>%
  left_join(tram_RAI_evenness_simpson_binned, by = "age") %>%
  left_join(tram20_SAR %>%
    dplyr::select(age, `Facultative coprophilous`, `Obligate coprophilous`), by = "age") %>%
  rename("fac_coprophilous" = "Facultative coprophilous",
        "ob_coprophilous" = "Obligate coprophilous") %>%
  left_join(Mendukilo_binned, by = "age") %>%
  left_join(RAI_plants_groups_binned, by = "age") %>%
  left_join(tram20_PAR %>%
    dplyr::select(age, Abies, Fagus, Poaceae) %>%
    rename("PAR_Abies" = "Abies",
          "PAR_Fagus" = "Fagus",
          "PAR_Poaceae" = "Poaceae"), by = "age")
```

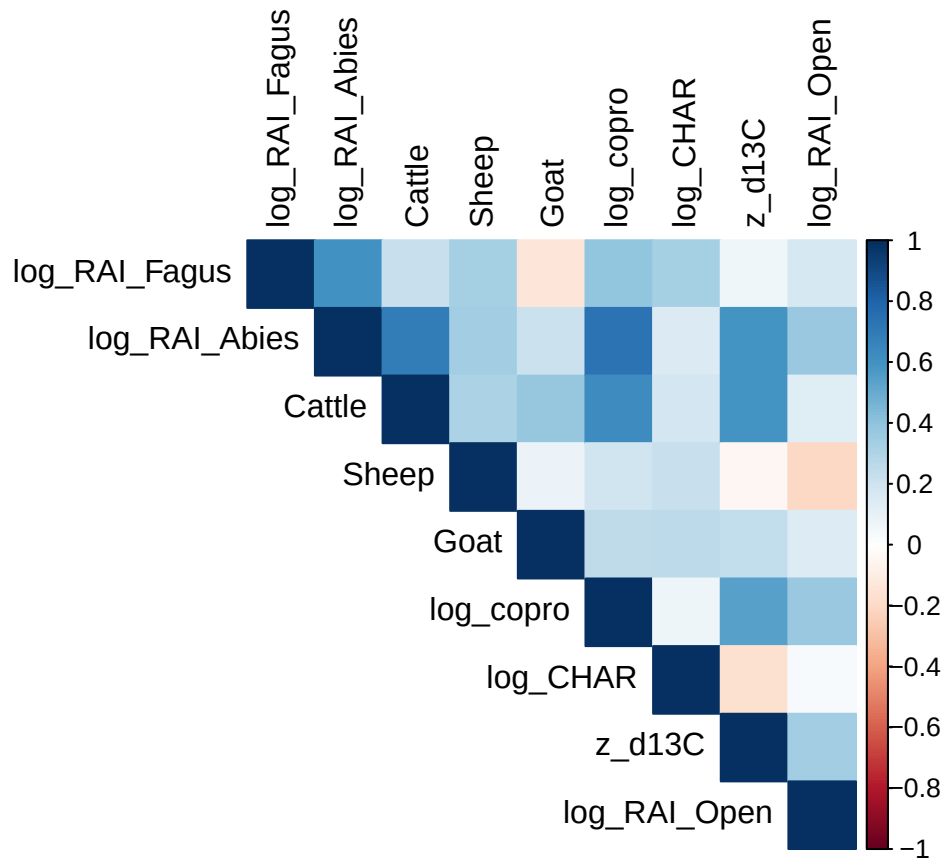
Transform variables as PAR and RAI have many 0s, and SEM assumes multivariate normality, so log-transformations improve normality, linearity and homocedasticity

```
data_transformed <- data_SEM %>%
  mutate(log_RAI_Abies = log10(RAI_Abies + 1e-7), # add small values for 0
        log_RAI_Fagus = log10(RAI_Fagus + 1e-7),
        log_RAI_Open = log10(RAI_Open + 1e-7),
        log_RAI_Novel = log10(RAI_Novel + 1e-7),
        log_PAR_Abies = log10(PAR_Abies + 1),
        log_PAR_Fagus = log10(PAR_Fagus + 1),
        log_PAR_Poaceae = log10(PAR_Poaceae + 1),
        log_CHAR = log10(Char.tram + 1),
        log_copro = log10(ob_coprophilous + 1),
        # standardise (optional, helps interpretation)
        z_d13C = as.numeric(scale(d13C.Mendukilo)),
        z_Cattle = as.numeric(scale(Cattle)),
        z_Sheep = as.numeric(scale(Sheep)),
        z_Goat = as.numeric(scale(Goat)),
        z_richness_RAI = as.numeric(scale(plant_richness_dna)),
        z_evenness_polen = as.numeric(scale(evenness_simpson)),
        z_evenness_RAI = as.numeric(scale(evenness_simpson_RAI)))
```

Explore correlations between variables

```
vars_cor <- data_transformed %>%
  dplyr::select(log_RAI_Fagus, log_RAI_Abies, Cattle, Sheep, Goat,
               log_copro, log_CHAR, z_d13C, log_RAI_Open)

cor_matrix <- cor(vars_cor, use = "complete.obs")
corrplot::corrplot(cor_matrix, method = "color",
                  type = "upper", tl.col = "black")
```



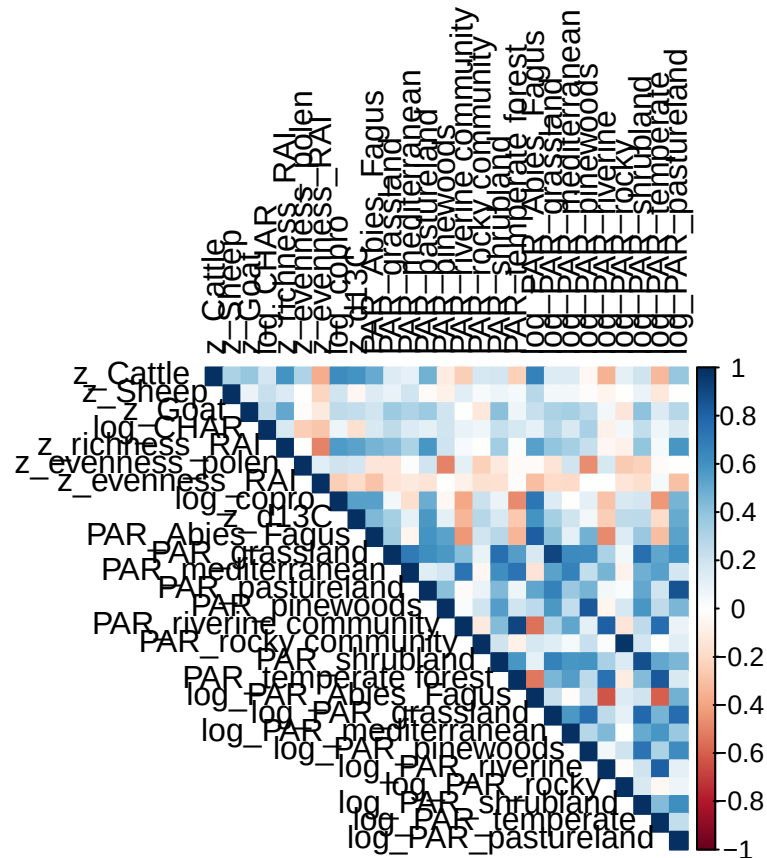
6.1. Prepare data for communities

```
data_SEM_communities <- data_transformed %>%
  dplyr::select(age, z_Cattle, z_Sheep, z_Goat, log_CHAR, z_richness_RAI, z_evenness_polen,
               z_evenness_RAI, log_copro, z_d13C) %>%
  left_join(tram20_PAR_groups_communities %>%
            rename_with(~ gsub("^", "PAR_", .x), .cols = Abies_Fagus:`temperate forest`),
            by = "age") %>%
  mutate(log_PAR_Abies_Fagus = log10(PAR_Abies_Fagus + 1), # add small values for 0
         log_PAR_grassland = log10(PAR_grassland + 1),
         log_PAR_mediterranean = log10(PAR_mediterranean + 1),
         log_PAR_pinewoods = log10(PAR_pinewoods + 1),
         log_PAR_riverine = log10(`PAR_riverine community` + 1),
         log_PAR_rocky = log10(`PAR_rocky community` + 1),
         log_PAR_shrubland = log10(PAR_shrubland + 1),
```

```
log_PAR_temperate = log10(`PAR_temperate forest` + 1),
log_PAR_pastureland = log10(PAR_pastureland + 1))
```

Explore correlations between variables

```
cor_matrix_communities <- cor(data_SEM_communities %>% dplyr::select(-age, -depth), use = "complete.obs")
corrplot::corrplot(cor_matrix_communities, method = "color",
  type = "upper", tl.col = "black")
```



6.2. Model with selected variables

```
tram_SEM_PAR_communities_selected <- '# Latent variables
Herbivory =~ NA*z_Cattle + z_Sheep + log_copro
Herbivory ~~ 1*Herbivory

# Structural regressions
# Effects of temperature (d13C) on disturbances
Herbivory ~ z_d13C

# Effects of disturbances on communities
log_PAR_Abies_Fagus ~ Herbivory
log_PAR_grassland ~ log_CHAR + z_d13C
log_PAR_pinewoods ~ log_CHAR
```

```

log_PAR_temperate ~ log_CHAR + Herbivory
log_PAR_mediterranean ~ z_d13C

#Effects of disturbances on heterogeneity
z_evenness_polen ~ Herbivory + log_CHAR

# Residual covariance between disturbances
Herbivory ~~ log_CHAR
,
fit_no_normal_8 <- cfa(tram_SEM_PAR_communities_selected, data = data_SEM_communities, test = "bootstrap")
summary(fit_no_normal_8, standardized = TRUE, fit.measures = TRUE, rsquare = TRUE)
fitMeasures(fit_no_normal_8, c("chisq", "df", "pvalue", "cfi", "tli", "rmsea", "srmr"))

```

See standardised effects (comparable)

```

parameters_signif_nonnormal8 <- parameterEstimates(fit_no_normal_8, standardized = TRUE) %>%
  filter(op == "~") %>%
  mutate(variable = paste(lhs, "<=", rhs),
         dir = ifelse(std.all > 0, "positive", "negative")) %>%
  dplyr::select(variable, est, std.all, dir, pvalue) %>%
  arrange(desc(abs(std.all)))

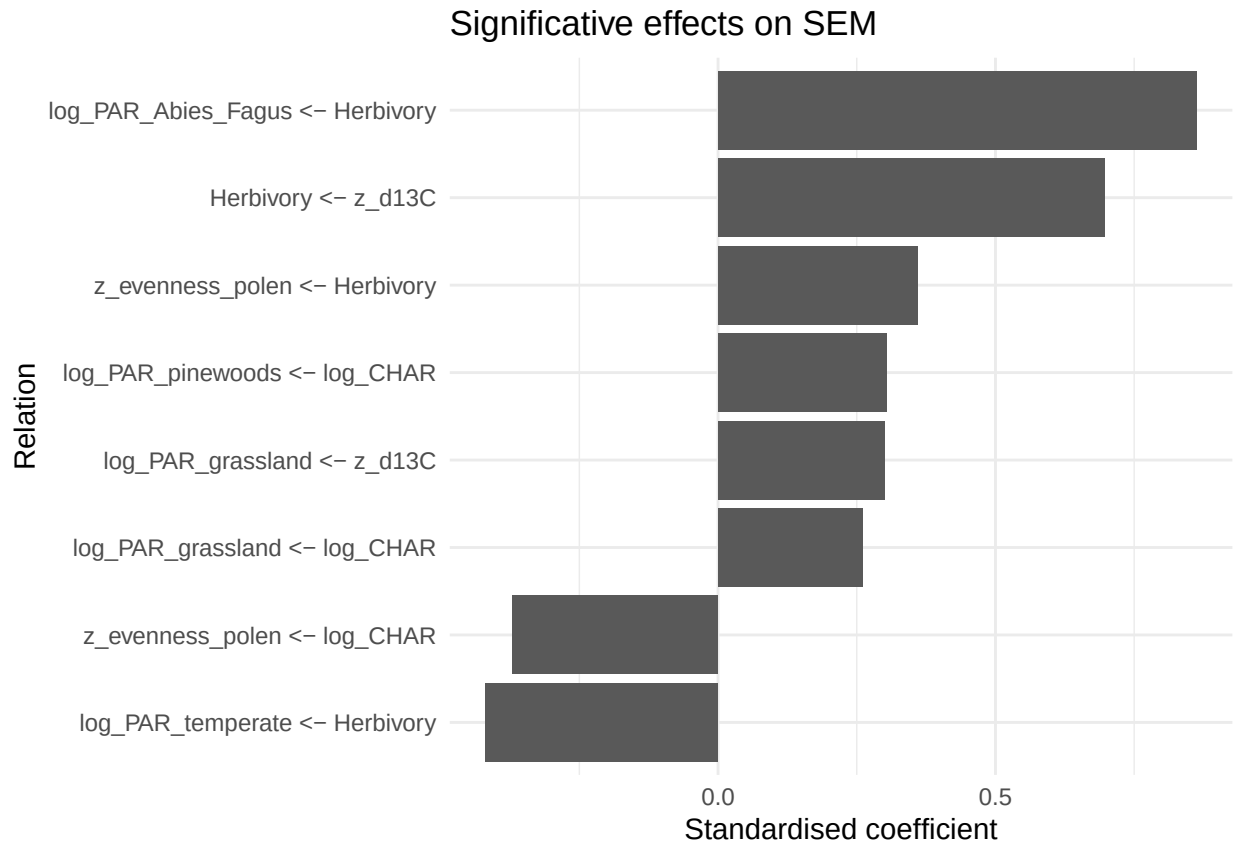
```

Figure of standardised coefficients

```

parameters_signif_nonnormal8 %>%
  filter(pvalue < 0.05) %>%
  ggplot(aes(x = reorder(variable, std.all), y = std.all)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(x = "Relation", y = "Standardised coefficient",
       title = "Significative effects on SEM") +
  theme_minimal()

```



Interpretation: Adequacy of the model -> NEARLY ADECUATE p-value = 0.463 CFI = 1 robust RMSEA = 0.002 SRMS = 0.077 Significant effects on Abies_Fagus community: Herbivory (+) Significant effects on herbivory: temperature (+) Significant effects on grasslands: temperature (+) fire (+) Significant effects on pinewoods: fire (+) Significant effects on temperate forest: herbivory (-) Significant effects on pollen evenness: herbivory (+) fire (-)

7. GAMs – Turnover vegetation drivers

7.1 Prepare data with all variables

```
bins_drivers <- scenario_4_RAI_peak$Age

Mendukilo_binned_drivers <- Mendukilo %>%
  mutate(binned_age = sapply(age, function(x) bins_drivers[which.min(abs(bins_drivers-x))])) %>%
  group_by(binned_age) %>%
  summarise(d13C.Mendukilo = mean(d13C.Mendukilo, na.rm = TRUE)) %>%
  rename("age" = "binned_age") %>%
  ungroup() %>%
  as.data.frame()

CHAR_binned_drivers <- CHAR %>%
  mutate(binned_age = sapply(age, function(x) bins_drivers[which.min(abs(bins_drivers-x))])) %>%
  group_by(binned_age) %>%
```

```

summarise(Char.tram = mean(Char.tram, na.rm = TRUE)) %>%
rename("age" = "binned_age") %>%
ungroup() %>%
as.data.frame()

RAI_animals_domestic_binned_drivers <- RAI_animals_domestic %>%
mutate(binned_age = sapply(age, function(x) bins_drivers[which.min(abs(bins_drivers-x))])) %>%
group_by(binned_age) %>%
summarise(Cattle = mean(Cattle, na.rm = TRUE),
          Sheep = mean(Sheep, na.rm = TRUE),
          Goat = mean(Goat, na.rm = TRUE)) %>%
rename("age" = "binned_age") %>%
ungroup() %>%
as.data.frame()

tram20_copro_binned_drivers <- tram20_SAR %>%
dplyr::select(age, `Facultative coprophilous`, `Obligate coprophilous`) %>%
rename("fac_coprophilous" = "Facultative coprophilous",
       "ob_coprophilous" = "Obligate coprophilous") %>%
mutate(binned_age = sapply(age, function(x) bins_drivers[which.min(abs(bins_drivers-x))])) %>%
group_by(binned_age) %>%
summarise(fac_coprophilous = mean(fac_coprophilous, na.rm = TRUE),
          ob_coprophilous = mean(ob_coprophilous, na.rm = TRUE)) %>%
rename("age" = "binned_age") %>%
ungroup() %>%
as.data.frame()

tram20_PAR_sum_binned_drivers <- tram20_PAR_sum %>%
dplyr::select(age, sum) %>%
mutate(binned_age = sapply(age, function(x) bins_drivers[which.min(abs(bins_drivers-x))])) %>%
group_by(binned_age) %>%
summarise(PAR_sum = mean(sum, na.rm = TRUE)) %>%
rename("age" = "binned_age") %>%
ungroup() %>%
as.data.frame()

scenario_4_PAR_peak_binned <- scenario_4_PAR_peak %>%
dplyr::select(Age, ROC) %>%
rename("age" = "Age") %>%
mutate(binned_age = sapply(age, function(x) bins_drivers[which.min(abs(bins_drivers-x))])) %>%
group_by(binned_age) %>%
summarise(ROC = mean(ROC, na.rm = TRUE)) %>%
rename("age" = "binned_age") %>%
ungroup() %>%
as.data.frame()

data_drivers_turnover <- scenario_4_RAI_peak %>%
dplyr::select(Age, ROC) %>%
rename("age" = "Age",
       "ROC_RAI" = "ROC") %>%
left_join(scenario_4_PAR_peak_binned %>%
          rename("ROC_PAR" = "ROC"), by = "age") %>%
left_join(Mendukilo_binned_drivers, by = "age") %>%

```

```

left_join(Char_binned_drivers, by = "age") %>%
left_join(RAI_animals_domestic_binned_drivers, by = "age") %>%
left_join(tram20_copro_binned_drivers, by = "age") %>%
left_join(tram20_PAR_sum_binned_drivers, by = "age")

data_drivers_turnover_transformed <- data_drivers_turnover %>%
  mutate(z_ROC_RAI = as.numeric(scale(ROC_RAI)),
         z_ROC_PAR = as.numeric(scale(ROC_PAR)),
         z_d13C = as.numeric(scale(d13C.Mendukilo)),
         z_Cattle = as.numeric(scale(Cattle)),
         z_Sheep = as.numeric(scale(Sheep)),
         z_Goat = as.numeric(scale(Goat)),
         z_CHAR = as.numeric(Char.tram),
         log_ob_copro = log10(ob_coprophilous + 1),
         log_fac_copro = log10(fac_coprophilous + 1),
         log_PAR = log10(PAR_sum + 1)) %>%
  drop_na()

```

7.2. Individual GAMs

See the effect of temperature on Rate of Change of sedaDNA

```

gam_drivers_1 <- gam(z_ROC_RAI ~ z_d13C, data = data_drivers_turnover_transformed)
summary_gam1 <- summary(gam_drivers_1) #R2 = 41% and significant

```

See the effect of temperature on Rate of Change of pollen

```

gam_drivers_1_PAR <- gam(z_ROC_PAR ~ z_d13C, data = data_drivers_turnover_transformed)
summary_gam1_PAR <- summary(gam_drivers_1_PAR) #R2 = 45% and significant

```

See the effect of herbivory (cattle) on Rate of Change of sedaDNA

```

gam_drivers_2 <- gam(z_ROC_RAI ~ z_Cattle, data = data_drivers_turnover_transformed)
summary_gam2 <- summary(gam_drivers_2) #R2 = 25% and significant

```

See the effect of herbivory (cattle) on Rate of Change of pollen

```

gam_drivers_2_PAR <- gam(z_ROC_PAR ~ z_Cattle, data = data_drivers_turnover_transformed)
summary_gam2_PAR <- summary(gam_drivers_2_PAR) #R2 = 33% and significant

```

See the effect of herbivory (Sheep) on Rate of Change of sedaDNA

```

gam_drivers_3 <- gam(z_ROC_RAI ~ z_Sheep, data = data_drivers_turnover_transformed)
summary_gam3 <- summary(gam_drivers_3) #R2 = 1% and not significant

```

See the effect of herbivory (Sheep) on Rate of Change of pollen

```

gam_drivers_3_PAR <- gam(z_ROC_PAR ~ z_Sheep, data = data_drivers_turnover_transformed)
summary_gam3_PAR <- summary(gam_drivers_3_PAR) #R2 = 1% and not significant

```

See the effect of herbivory (coprophilous fungi) on Rate of Change of sedaDNA

```
gam_drivers_4 <- gam(z_ROC_RAI ~ log_ob_copro, data = data_drivers_turnover_transformed)
summary_gam4 <- summary(gam_drivers_4) #R2 = 47% and significant
```

See the effect of herbivory (coprophilous fungi) on Rate of Change of pollen

```
gam_drivers_4_PAR <- gam(z_ROC_PAR ~ log_ob_copro, data = data_drivers_turnover_transformed)
summary_gam4_PAR <- summary(gam_drivers_4_PAR) #R2 = 25% and significant
```

See the effect of fire on Rate of Change of sedaDNA

```
gam_drivers_5 <- gam(z_ROC_RAI ~ z_CHAR, data = data_drivers_turnover_transformed)
summary_gam5 <- summary(gam_drivers_5) #R2 = 0.05% and not significant
```

See the effect of fire on Rate of Change of pollen

```
gam_drivers_5_PAR <- gam(z_ROC_PAR ~ z_CHAR, data = data_drivers_turnover_transformed)
summary_gam5_PAR <- summary(gam_drivers_5_PAR) #R2 = 0.2% and not significant
```